

Objective function

Differentiability: the first order

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Optimization: principles and algorithms

Differentiability: the first order

Derivatives play a central role in optimization. Let's recall some basic definitions that are relevant for our purposes.

Directional derivative

$f : \mathbb{R}^n \rightarrow \mathbb{R}$ continuous

$$\nabla f(x)^T d = \lim_{\substack{\alpha > 0 \\ \alpha \rightarrow 0}} \frac{f(x + \alpha d) - f(x)}{\alpha},$$

if the limit exists.

Differentiable function

f such that $\nabla f(x)^T d$ exists for any $d \in \mathbb{R}^n$.

Partial derivative

$$\frac{\partial f}{\partial x_i} = \nabla f(x)^T e_i$$

where

$$e_i = (0, 0, \dots, 1, \dots, 0)$$

Gradient

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f(x)}{\partial x_1} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{pmatrix} \in \mathbb{R}^n$$

Descent direction

d is a descent direction in x for f if

$$d^T \nabla f(x) < 0.$$

Generalization to several functions

$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ continuous

Gradient matrix

$$\begin{aligned}\nabla f(x) &= \begin{pmatrix} \left. \nabla f_1(x) \right| & \cdots & \left. \nabla f_m(x) \right| \\ \hline \end{pmatrix} \\ &= \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_2}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_1} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial f_1}{\partial x_n} & \frac{\partial f_2}{\partial x_n} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}.\end{aligned}$$

Jacobian matrix

$$\begin{aligned}J(x) &= \nabla f(x)^T \\ &= \begin{pmatrix} \text{---} & \nabla f_1(x)^T & \text{---} \\ & \vdots & \\ \text{---} & \nabla f_m(x)^T & \text{---} \end{pmatrix}.\end{aligned}$$

Summary

We will exploit the derivatives both for the analysis of the problem and the algorithms.

In this course, we assume that the objective function is differentiable.